

FINAL PROJECT REPORT

E-Commerce Customer Segmentation & Predictive Behavior Modeling

Data-Driven Segmentation, Machine Learning Classification, and Actionable Retention Strategy

Prepared for: Executive Leadership & Marketing Stakeholders

Prepared by: Data Science & Analytics Team

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Executive Summary

This report presents the findings and business outcomes of the E-Commerce Customer Segmentation and Prediction project, undertaken to strengthen marketing strategy and improve customer retention through data-driven decision-making. Using transactional data from 541,909 order line items, the project applied rigorous data cleaning, RFM (Recency, Frequency, Monetary) feature engineering, unsupervised clustering, and supervised classification to translate raw purchase history into a segmentation framework the business can act on immediately.

After cleaning, 392,692 valid transaction records were retained and aggregated into customer-level RFM profiles. K-Means clustering ($k = 4$) was selected as the optimal segmentation model based on its silhouette performance, computational scalability, and the interpretability of the resulting groups. Four distinct personas emerged: Champions, Loyal/Regulars, New/Promising, and Lost/Churned customers.

To operationalize these segments for real-time and future customers, a Random Forest classifier was trained and tuned via GridSearchCV, achieving 99% accuracy in predicting a customer's segment from their RFM behavior. Recency was identified as the dominant predictor of segment membership, followed by Frequency and Monetary value — a finding with direct implications for how and when marketing teams should intervene.

The combined segmentation and prediction framework equips the business to move from generic, one-size-fits-all campaigns to precision-targeted marketing. Key anticipated impacts include:

- **Enhanced Customer Segmentation:** Marketing can now target customers based on validated behavioral groups rather than broad demographic assumptions.
- **Cost Efficiency:** Retention and acquisition budgets can be reallocated toward the highest-value and highest-risk segments, reducing spend waste on low-propensity audiences.
- **Improved Customer Satisfaction:** Personalized outreach — tailored to each persona's actual purchasing behavior — improves relevance and strengthens customer relationships.

The remainder of this report details the methodology, model evaluation, persona profiles, feature importance analysis, business insights, limitations, and specific, ready-to-implement marketing recommendations for each customer segment.

1. Introduction

1.1 Business Context

E-commerce businesses generate large volumes of transactional data, yet this data is frequently underutilized for strategic decision-making. Without a systematic way to distinguish high-value customers from at-risk or low-engagement ones, marketing efforts tend to be broad, inefficient, and reactive rather than targeted and proactive.

1.2 Project Objectives

This project was commissioned to answer two core business questions:

- Who are our customers, and how do their purchasing behaviors differ from one another?
- Can we reliably predict which behavioral segment a customer belongs to — or is heading toward — so marketing action can be taken proactively?

To address these questions, the project pursued the following technical objectives:

- Clean and prepare raw transactional order data for analysis.

- Engineer Recency, Frequency, and Monetary (RFM) features that quantify customer purchasing behavior.
- Apply unsupervised clustering to discover natural customer segments.
- Build and tune a supervised classification model to predict segment membership for new or existing customers.
- Translate model outputs into concrete marketing recommendations.

1.3 Strategic Goals

The overarching business goals of this initiative are to enhance customer segmentation precision, improve marketing cost efficiency, and increase customer satisfaction and retention through more relevant, timely engagement — directly supporting revenue stability and long-term customer lifetime value.

2. Data Methodology

2.1 Data Cleaning

The raw dataset comprised 541,909 transactional line items. A structured cleaning pipeline was applied to ensure the analysis reflects genuine, attributable customer purchases:

Stage	Description	Rows Remaining
Raw dataset	Original transactional order line items	541,909
Missing Customer IDs removed	Dropped records lacking a CustomerID; these cannot be attributed to an individual customer and cannot support RFM aggregation	406,829
Invalid transactions & duplicates removed	Removed rows with negative Quantity or negative/zero UnitPrice (product returns and data-entry errors), and dropped exact duplicate transaction rows	392,692

Table 1. Data cleaning pipeline from raw transactions to final analytical dataset.

- **Missing CustomerIDs:** 135,080 rows lacked a CustomerID and were removed, as these transactions — largely guest checkouts — cannot be tied to a unique customer for RFM analysis.
- **Returns and data errors:** Rows with negative Quantity (product returns) or non-positive UnitPrice (cancellations, adjustments, or data-entry errors) were excluded, as they do not represent completed, revenue-generating purchases.
- **Duplicate rows:** Exact duplicate records were dropped to avoid artificially inflating a customer's purchase frequency or monetary value.
- **Result:** The cleaning pipeline reduced the dataset to 392,692 valid transaction line items, forming a reliable foundation for customer-level feature engineering.

2.2 RFM Feature Engineering

Transactional (order-line) data is not directly usable for customer segmentation — it must first be aggregated to the customer level. This was accomplished using the RFM framework, an industry-standard approach in retail and e-commerce analytics:

- **Recency (R):** Number of days since the customer's most recent purchase, relative to a fixed analysis (snapshot) date. Lower recency indicates a more recently active customer.

- **Frequency (F):** Total number of distinct purchase transactions (invoices) made by the customer. Higher frequency indicates a more habitual, engaged buyer.
- **Monetary (M):** Total revenue generated by the customer across all transactions. Higher monetary value indicates greater cumulative spend and revenue contribution.

Each of the 392,692 cleaned transaction rows was grouped by CustomerID and aggregated into these three metrics, producing one RFM record per unique customer.

2.3 Outlier Treatment and Scaling

Monetary value in retail datasets is typically right-skewed, with a small number of exceptionally large orders (e.g., bulk or wholesale purchases). Left untreated, these extreme values would disproportionately pull K-Means cluster centroids toward a handful of outliers, distorting the segmentation for the broader customer base.

- **IQR-based outlier filtering:** The Interquartile Range (IQR) method was applied to the Monetary feature, removing values falling outside 1.5x the IQR above the third quartile, to prevent centroid skewing during clustering.
- **Standardization:** Following outlier treatment, StandardScaler was applied to the RFM feature set, rescaling Recency, Frequency, and Monetary to zero mean and unit variance. This step is essential for K-Means, which relies on Euclidean distance and would otherwise be dominated by whichever feature has the largest raw numeric scale (typically Monetary).

The resulting scaled RFM matrix served as the direct input to the clustering models described in Section 3.

3. Unsupervised Clustering (Customer Segmentation)

3.1 Model Comparison

Three clustering algorithms were evaluated on the scaled RFM feature set to determine the most effective approach for segmenting the customer base:

Model	Approach	Key Consideration	Verdict
K-Means (k=4)	Centroid-based partitioning into a pre-specified number of clusters	High silhouette score; fast and scalable on the full customer base; produced well-balanced, interpretable segments	Selected
Hierarchical Clustering	Agglomerative merging of customers based on a linkage criterion, visualized via dendrogram	Computationally expensive at scale; less practical for repeated re-scoring of new customers	Not selected
DBSCAN	Density-based clustering that groups closely packed points and flags outliers as noise	Sensitive to eps/min_samples tuning; produced uneven cluster sizes and a large noise segment on RFM data	Not selected

Table 2. Comparative evaluation of clustering algorithms on the RFM feature set.

3.2 Justification for K-Means (k = 4)

K-Means with k = 4 was selected as the final segmentation model for three primary reasons:

- **Cluster quality:** K-Means achieved the highest Silhouette Score among the three approaches, indicating well-separated, internally cohesive clusters.
- **Scalability:** As a centroid-based algorithm, K-Means scales efficiently to the full customer base and can be re-applied cheaply as new transactional data arrives.
- **Interpretability and balance:** The four resulting clusters produced business-relevant, reasonably balanced group sizes that map cleanly onto actionable marketing personas — unlike DBSCAN's uneven, noise-heavy output.

The value $k = 4$ was further supported by standard cluster-count diagnostics (e.g., elbow-method inertia reduction and silhouette comparison across candidate k values), which favored four clusters as the point of diminishing returns in intra-cluster variance reduction.

3.3 Customer Persona Breakdown

Each of the four clusters was profiled against its average Recency, Frequency, and Monetary characteristics and mapped to a named, business-friendly persona:

Persona	Customer Count	Behavioral Profile
Champions	355	Most recent purchases, highest purchase frequency, and highest monetary value. The most valuable and engaged customer group.
Loyal / Regulars	1,741	Consistent, moderately frequent purchasers with solid monetary contribution. The largest segment and the backbone of recurring revenue.
New / Promising	862	Recent first-time or early-stage buyers with low frequency and monetary value to date. High growth potential if nurtured.
Lost / Churned	955	Long recency (extended time since last purchase), with historically low-to-moderate frequency and monetary value. At high risk of permanent attrition.

Table 3. Final customer segments and population sizes.

Together, these four personas provide a complete, mutually exclusive view of the active customer base, ranked implicitly by engagement and value — from Champions at the top to Lost/Churned customers requiring urgent re-engagement.

4. Predictive Modeling

While clustering identifies segments retrospectively, the business also needs to classify customers — including new or recently active ones — into these same personas going forward, without re-running the full clustering pipeline. To this end, the K-Means cluster labels were used as the target variable for a supervised classification task.

4.1 Model Selection

Two classification algorithms were trained and compared on the RFM feature set, using the K-Means-derived persona labels as ground truth:

Model	Role	Accuracy
Logistic Regression	Baseline linear classifier used as a performance benchmark	Baseline

Random Forest (tuned)	Ensemble tree-based classifier, hyperparameter-tuned via GridSearchCV	99%
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Table 4. Classifier comparison and final selected model.

Logistic Regression served as an interpretable linear baseline. The Random Forest classifier substantially outperformed it, owing to its ability to capture non-linear relationships and interaction effects between Recency, Frequency, and Monetary value — which do not separate customer segments along purely linear boundaries.

4.2 Hyperparameter Tuning

The Random Forest model was optimized using GridSearchCV, an exhaustive cross-validated search over a defined hyperparameter grid, to identify the configuration that best balances predictive accuracy with generalization (avoiding overfitting):

Hyperparameter	Tuned Value	Purpose
n_estimators	50	Number of decision trees in the ensemble
max_depth	10	Maximum depth per tree, controlling overfitting
min_samples_split	5	Minimum samples required to split an internal node
min_samples_leaf	1	Minimum samples required at a leaf node

Table 5. Final tuned Random Forest hyperparameters.

4.3 Evaluation Metrics

- **Accuracy:** The tuned Random Forest classifier achieved 99% accuracy on the held-out test set, confirming that a customer's RFM profile is a highly reliable predictor of their behavioral segment.
- **Confusion Matrix:** Class-by-class evaluation showed the model correctly assigned the overwhelming majority of customers to their true persona, with minimal misclassification concentrated between adjacent segments (e.g., Loyal/Regulars vs. Champions), where behavioral boundaries are naturally closest.
- **ROC-AUC:** Per-class ROC-AUC scores were consistently high across all four personas, indicating strong separability and confirming that the model's predictions are reliable enough to drive automated, real-time marketing triggers rather than requiring manual review.

The near-ceiling accuracy is consistent with the fact that the target labels were themselves derived from the RFM features via K-Means — meaning the classifier's task is to learn the geometric decision boundaries K-Means already established. This makes the model an efficient, production-ready way to score new or existing customers into personas without re-running clustering on the entire customer base each time.

5. Feature Importance

Analysis of the Random Forest's internal feature importance scores reveals a clear hierarchy in what drives customer segment membership:

Rank	Feature	Relative Influence on Segment Prediction
1	Recency	Dominant predictor — how recently a customer purchased is the single strongest signal of which segment they belong to.
2	Frequency	Second most influential — the number of past transactions strongly

		reinforces segment separation, particularly between Champions and Loyal/Regulars.
3	Monetary	Contributes meaningfully but less than Recency or Frequency — total spend refines segment assignment rather than driving it outright.

Table 6. Feature importance ranking from the tuned Random Forest classifier.

5.1 Interpretation

The dominance of Recency has a direct and important business implication: how recently a customer last purchased is a more powerful signal of their current relationship with the brand than how much they have spent historically or how often they have bought in the past. In practical terms, this means:

- A customer's segment status can shift quickly based on their most recent activity, making Recency the ideal trigger for time-sensitive marketing automation (e.g., win-back campaigns, re-engagement emails).
- Frequency provides confirmatory signal — it distinguishes habitual buyers from one-time purchasers within similar Recency bands.
- Monetary value, while still relevant, plays a supporting role — high past spend alone does not protect a customer from being reclassified as Lost/Churned if Recency deteriorates.

This ranking should directly inform the cadence and prioritization of retention campaigns: recency-based triggers should take precedence over spend-based ones when deciding who to target and when.

6. Business Insights & Limitations

6.1 Key Business Insights

- **Retention risk is concentrated:** 955 customers (the Lost/Churned segment) represent a substantial pool of prior revenue that has gone dormant — a clear, quantified target for win-back investment.
- **The loyal base is the revenue backbone:** At 1,741 customers, Loyal/Regulars form the largest segment, underscoring the importance of retention and satisfaction programs over pure acquisition spend.
- **Champions are disproportionately valuable but few:** At only 355 customers, Champions likely account for an outsized share of total revenue, warranting white-glove retention treatment.
- **New/Promising customers are a conversion opportunity:** 862 customers are early in their lifecycle; how they are nurtured in the coming weeks will determine whether they progress toward Loyal/Champion status or drift toward churn.
- **Segment membership is predictable and automatable:** With 99% classification accuracy, the business can confidently automate segment scoring in near real time rather than relying on periodic manual analysis.

6.2 Model & Data Limitations

- **K-Means distance sensitivity:** K-Means relies on Euclidean distance to spherical cluster assumptions; it can be sensitive to residual outliers and may not capture non-convex or highly irregular segment shapes in customer behavior.
- **Dropped guest checkouts:** 135,080 rows without a CustomerID — largely guest checkouts — were excluded from the analysis. This population's behavior is entirely invisible to the segmentation model, meaning the framework only reflects identifiable, returning-customer behavior.
- **Circular evaluation on classifier accuracy:** Because the Random Forest was trained on labels generated by K-Means itself, its 99% accuracy reflects how well it learned the clustering boundaries — not an independent, ground-truth validation of segment correctness.

- **Static snapshot:** RFM features are computed relative to a fixed snapshot date. Without a refresh cadence, segment assignments will grow stale as customer behavior evolves.
- **Single-market, single-period scope:** The underlying dataset reflects a specific transactional period and customer base; seasonality, promotions, or regional variation are not explicitly modeled and may affect generalizability.
- **Outlier exclusion trade-off:** IQR filtering on Monetary value, while necessary to protect centroid integrity, also excludes legitimate high-value/bulk buyers from the clustering input — these customers should be separately reviewed rather than dropped from business consideration entirely.

7. Actionable Recommendations

The following persona-specific strategies translate the segmentation and prediction results into concrete marketing actions, prioritized by expected business impact.

Champions (355 customers)

Highest-value, most engaged customers — protect and reward this relationship above all else.

- Enroll automatically in a top-tier VIP loyalty program with exclusive early access to new product launches.
- Assign priority customer service and expedited shipping to reinforce their premium experience.
- Deploy personalized, high-touch outreach (e.g., handwritten-style thank-you notes, milestone rewards) rather than mass promotions.
- Use as a source for referral and advocacy programs — incentivize reviews and referrals given their high brand affinity.

Loyal / Regulars (1,741 customers)

The largest and most stable segment — the core revenue engine that should be nurtured toward Champion status.

- Implement tiered loyalty incentives (points, cashback, or milestone discounts) that reward increasing purchase frequency.
- Cross-sell and upsell using purchase-history-based product recommendations to increase basket size.
- Maintain consistent, moderate-cadence communication (e.g., monthly newsletters, restock alerts) to sustain engagement without over-messaging.
- Monitor Recency closely per Section 5 — early drift toward longer gaps between purchases should trigger proactive re-engagement before this group risks churning.

New / Promising (862 customers)

Early-lifecycle customers with high growth potential — the critical window for building loyalty habits.

- Launch a structured onboarding journey (welcome series, second-purchase incentive) within the first 30–60 days of first purchase.
- Offer a limited-time second-purchase discount to accelerate the transition from one-time buyer to repeat customer.
- Use product-education content to build category familiarity and encourage exploration beyond the first purchase.
- Track this cohort's Frequency trend closely, since increasing purchase frequency is the clearest early indicator of progression toward the Loyal/Regulars segment.

Lost / Churned (955 customers)

Long-dormant customers at high risk of permanent attrition — requires a distinct win-back strategy, not standard promotions.

- Deploy a dedicated win-back campaign with a meaningful incentive (e.g., time-limited discount or free shipping) rather than routine marketing emails.
- Survey a sample of this segment to understand churn drivers (price, competition, service experience) and feed findings back into product/marketing strategy.
- Reduce marketing spend on repeated generic promotions to this group where response history shows low engagement; reallocate budget toward Loyal/Regulars and New/Promising segments.
- For customers with high historical Monetary value prior to churn, prioritize personal outreach (e.g., account manager check-in) over automated email, given their higher potential recovery value.

7.1 Cross-Segment Priorities

- **Automate segment scoring:** Integrate the tuned Random Forest model into the CRM or marketing automation platform to score customers into personas on a recurring (e.g., weekly) basis, enabling near real-time campaign targeting.
- **Prioritize Recency-based triggers:** Given Recency's dominant predictive weight, configure lifecycle marketing automation to react primarily to changes in purchase recency rather than cumulative spend alone.
- **Establish a refresh cadence:** Re-run RFM aggregation and re-score the customer base on a regular schedule (e.g., monthly) to keep segment assignments current as behavior evolves.
- **Address the guest-checkout blind spot:** Evaluate incentivizing account creation at checkout to reduce the volume of unattributable transactions and extend segmentation coverage in future iterations.